EXPERIMENT 10

-==========
-- Name: VANSHIKA DIXIT
-- Roll No: 2401010010
-- Topic: JOINS & SUBQUERY
-==========

Using the tables "DEPARTMENTS" and "EMPLOYEES" perform the following queries

a) Display the employee details, departments that the departments are same in both the emp and dept.

b) Display the employee name and Department name by implementing a left outer join.

c) Display the employee name and Department name by implementing a right outer join.

d) Display the details of those who draw the salary greater than the average salary

```
CREATE DATABASE IF NOT EXISTS Assignment10_DB;

USE Assignment10_DB;


CREATE TABLE Departments (

   dept_no INT PRIMARY KEY,

   dept_name VARCHAR(50),

   dept_location VARCHAR(50)

);


CREATE TABLE Employees (

   emp_id INT PRIMARY KEY,

   emp_name VARCHAR(50),

   emp_salary DECIMAL(10,2),

   dept_no INT

);


INSERT INTO Departments VALUES

(1, 'HR', 'Delhi'),
```

(2, 'Finance', 'Mumbai'),

(3, 'IT', 'Bangalore');

INSERT INTO Employees VALUES

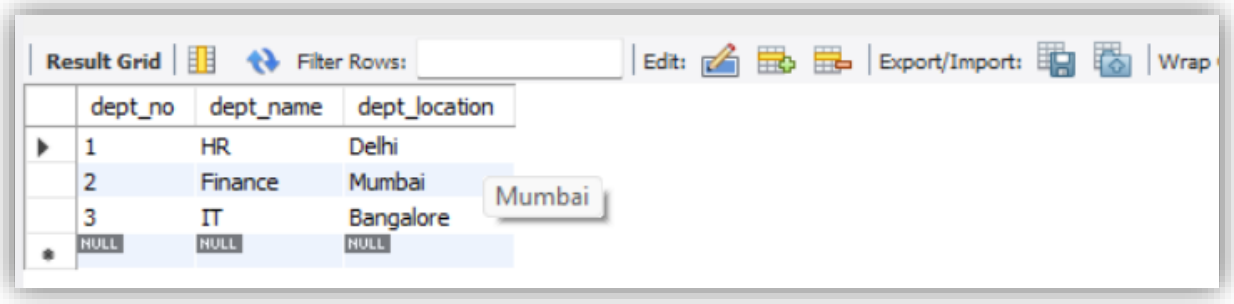
(101, 'Vanshika Dixit', 40000, 1),

(102, 'Roshani Singh', 50000, 2),

(103, 'Aman', 60000, 3),

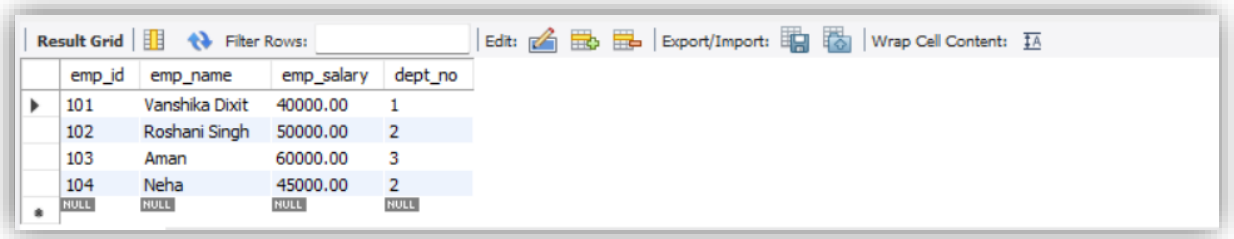
(104, 'Neha', 45000, 2);

SELECT * FROM Departments;



dept_no	dept_name	dept_location
1	HR	Delhi
2	Finance	Mumbai
3	IT	Bangalore
NULL	NULL	NULL

SELECT * FROM Employees;



emp_id	emp_name	emp_salary	dept_no
101	Vanshika Dixit	40000.00	1
102	Roshani Singh	50000.00	2
103	Aman	60000.00	3
104	Neha	45000.00	2
NULL	NULL	NULL	NULL

SELECT *
FROM Employees e
INNER JOIN Departments d
ON e.dept_no = d.dept_no;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
101	Vanshika Dixit	40000.00	1	1	HR	Delhi
102	Roshani Singh	50000.00	2	2	Finance	Mumbai
103	Aman	60000.00	3	3	IT	Bangalore
104	Neha	45000.00	2	2	Finance	Mumbai

```
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.dept_no = d.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Vanshika Dixit	HR
Roshani Singh	Finance
Aman	IT
Neha	Finance

```
SELECT e.emp_name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d
ON e.dept_no = d.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Vanshika Dixit	HR
Neha	Finance
Roshani Singh	Finance
Aman	IT

```
SELECT *
FROM Employees
WHERE emp_salary > (SELECT AVG(emp_salary) FROM Employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no
102	Roshani Singh	50000.00	2
103	Aman	60000.00	3
NULL	NULL	NULL	NULL

FULL HISTORY

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:39:47	CREATE DATABASE IF NOT EXISTS Assignment10_DB	1 row(s) affected	0.031 sec
2	10:39:47	USE Assignment10_DB	0 row(s) affected	0.000 sec
3	10:39:47	CREATE TABLE Departments (dept_no INT PRIMARY KEY, dept_name VARCHAR(50), dept_location VAR...	0 row(s) affected	0.031 sec
4	10:39:47	CREATE TABLE Employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), emp_salary DECIMAL(...	0 row(s) affected	0.031 sec
5	10:39:47	INSERT INTO Departments VALUES (1, 'HR', 'Delhi'), (2, 'Finance', 'Mumbai'), (3, 'IT', 'Bangalore')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
6	10:40:35	INSERT INTO Employees VALUES (101, 'Vanshika Dixit', 40000, 1), (102, 'Roshani Singh', 50000, 2), (103, 'Aman', 6...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec
7	10:40:35	SELECT * FROM Departments LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
8	10:43:22	SELECT * FROM Employees LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec
9	10:43:59	SELECT * FROM Employees e INNER JOIN Departments d ON e.dept_no = d.dept_no LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
10	10:44:57	SELECT e.emp_name, d.dept_name FROM Employees e LEFT JOIN Departments d ON e.dept_no = d.dept_no LIMI...	4 row(s) returned	0.016 sec / 0.000 sec
11	10:45:35	SELECT e.emp_name, d.dept_name FROM Employees e RIGHT JOIN Departments d ON e.dept_no = d.dept_no LI...	4 row(s) returned	0.016 sec / 0.000 sec
12	10:46:15	SELECT * FROM Employees WHERE emp_salary > (SELECT AVG(emp_salary) FROM Employees) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec